31. THE GENITAL SYSTEM : IMPLANTATION

DURATION : 05:37 STUDY SHEET

COURSE SUMMARY

This video deeply explores the establishment of the genital system, focusing on the influence of the notochord and the nervous system on the development of genital structures. Key concepts include diaphragmatisation, the development of the adrenal glands and gonads, as well as the dynamic interaction between these organs, which guides the formation of the fallopian tubes, uterus, and Wolffian ducts. Through a biodynamic approach, practitioners will discover how these embryonic processes influence morphology and ligamentous structure, thus offering essential keys for an integrative understanding of the evolution of the urogenital systems in men and women.

THE GENITAL SYSTEM: ESTABLISHMENT

INFLUENCE OF THE NOTOCHORD AND NERVOUS SYSTEM

The **notochordal process** influences the **neural system**, which in turn impacts the **cardiac system**.

The cardiac system plays a crucial role in the establishment at the diaphragmatic level, more precisely at the level of the **transverse septum**, a phenomenon known as **diaphragmatisation**.

This diaphragmatisation process is also responsible for the development of the **adrenal glands**.

DEVELOPMENT OF ADRENALS AND GONADS

The growth of the **liver** generates a suction field, creating a posterior space in the **posterior parietal peritoneum**. This suction field contributes to the formation of the adrenals, which become very voluminous.

In parallel, the **urogenital system** is characterised by the migration of **gonocytes** from the **vitelline vesicle**. These gonocytes colonise the **genital ridge**, located at the posterior wall, thus stimulating the thickening of the epithelium and forming the **gonadal ridge**.

INTERACTION BETWEEN ADRENALS AND GONADS

The adrenals, pushed by the development of the liver, push the gonads laterally.

This developmental movement generates a **lateral fold** in the tissue, caused by the pressure of the adrenals on the gonads. This fold, visible in the posterior parietal peritoneum, is referred to as the **Müllerian duct**.

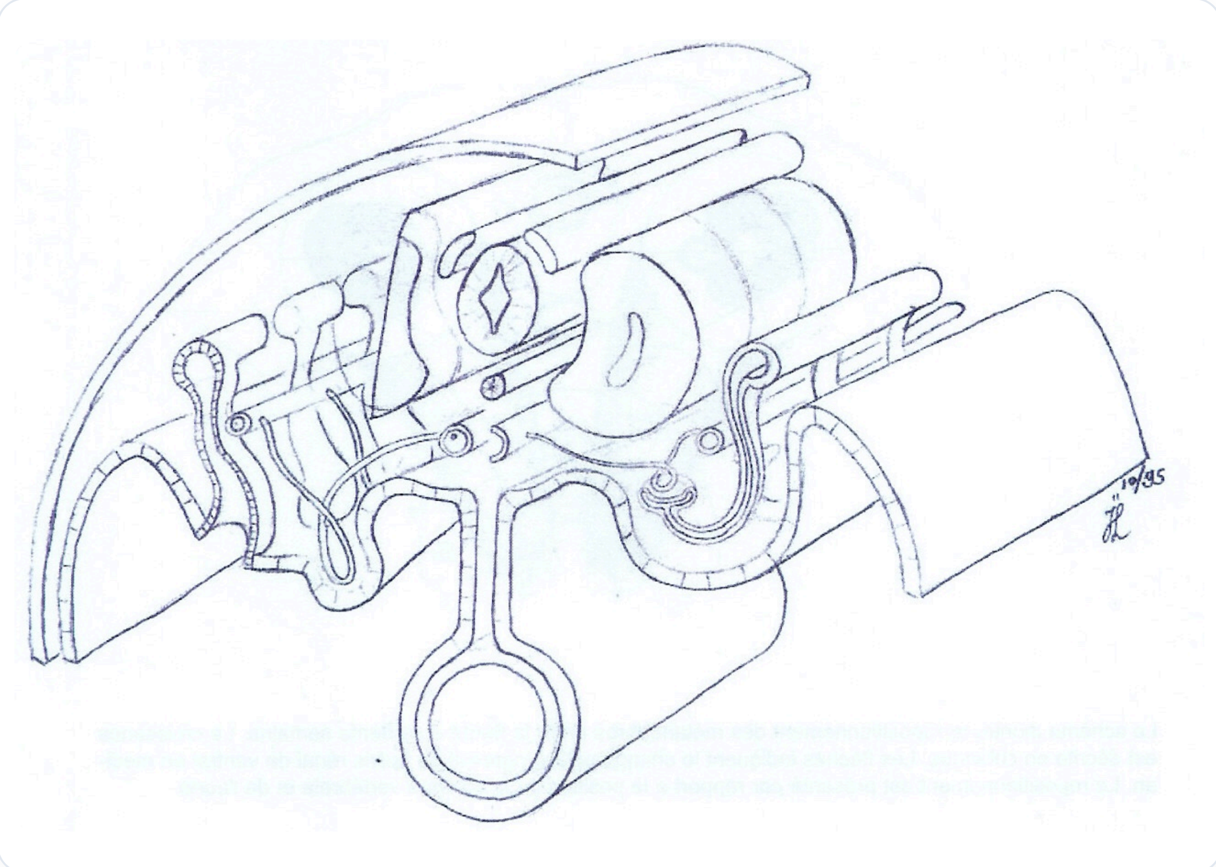

FIG. — This image seems to depict the effect of adrenals pushing gonads laterally, forming the Mullerian duct, as described

SEXUAL DIFFERENCES AND FORMATION OF GENITAL ORGANS

Initially, there is no visible structural difference between the sexes. However, genetically, females have more developed adrenals, accentuating the lateral movement.

This movement creates a fold in the tissue, which will give rise to the **Fallopian tubes** (left and right).

In the centre, the moving liver brings the elements closer together, while the lower limbs begin to form, creating a midline.

The two tubes of the Müllerian duct join, and under the influence of the liver and adrenals pushing the gonads laterally, this midline is structured.

Superiorly, the liver continues to exert traction, and the joining of the two tubes will lead to the formation of the **uterus** in females.

In males, this canal regresses and does not develop in the same way. The developing gonad will transform into an **ovary** in females, with the uterus originating from the posterior parietal peritoneum.

LIGAMENTOUS STRUCTURES AND DUCTS

The **broad ligament**, with its internal lines, will give rise to the **ovarian ligament**, the **uterosacral ligament**, and the **loose broad ligament**, all derived from the posterior parietal peritoneum.

The genital ridge, informed and developed, is linked to the development of the liver, which creates a new **retroperitoneal** space.

This suction field, typical of glands, pushes the gonads and kidneys, leading to the disappearance of the **mesonephros**.

The gonadal ridge itself moves laterally, forming a fold in the posterior parietal peritoneum, which will become the **Müllerian tubercle**.

The first level, originating from the collecting tubule, is the **mesonephric duct**, also known as the **Wolffian duct**. This duct will later serve as a conduit for the gonadal ridge (future testis or ovary).

Finally, the **diaphragmatic suspensory ligament** superiorly and the **gubernaculum** inferiorly will guide the ovaries into place in females or allow the descent of the testes through the **inguinal canal** in males.